

Armi & Millard (1976)

↳ Wimbush & Munk [1971], experiments
Weatherly [1972, 1975], Caldwell et al. [1972]
Howroyd & Stansbury [1975]

Turbulent Ekman layer $h_e = \frac{0.4 u_*}{f}$
Csanady [1967]
 $u_* = \frac{1}{30} U$
Empirical: $\frac{h_e}{U} \approx 2 \times 10^{-2} \text{ s}$
 $\frac{h}{U} \approx 1.2 \times 10^{-3} \text{ s}$
6x bigger

For $U \sim 4 \text{ cm s}^{-1}$, $h \sim 8 \text{ m}$ small.

↳ + high frequency motion (Wimbush + Munk [1971])

↳ Only contributes additional 6m $\rightarrow$ too small

(most energetic inertial/tidal velocities $\sim 3 \text{ cm s}^{-1}$.)

↳ + Stratification & rotation (Thompson [1973])

↳ Penetration height $\sim 8 \text{ m}$ for $U \sim 5 \text{ cm s}^{-1} \rightarrow$ too small

↳ Hatteras abyssal plane

Centre of HAP $\rightarrow$ structure characteristic of mixing uniformly stratified region

Rough/sloping topography show complicated structure

Edge of HAP $\rightarrow$ simple or complicated structure

↳ prelim. survey of HAP has

- spatial var. over 10km
- temporal var. over 3 days

Penetration height of well-mixed region correlated to mean velocity $\rightarrow$ dynamic origin

↳ "formation due to mixing stratified deep ocean"

13.1 Introduction

13.2 Vertical variation of density

Density stratification : $\frac{dp_0}{dz} = \frac{dp}{dz} + \frac{g\rho}{c^2}$ potential density gradient

Stable stratification if ρ_0 decreases with height.

① : in situ density change due to all sources (e.g. P, T, concentrations $\begin{matrix} \nearrow \text{salinity} \\ \searrow \text{water vapour} \end{matrix}$)

② : density gradient due to pressure decrease with height in adiabatic environment
 $\hookrightarrow$ adiabatic density gradient.

[can also have adiabatic temperature gradient]

e.g. Incompressible fluids : $c = \infty \Rightarrow$ zero adiabatic density gradient.

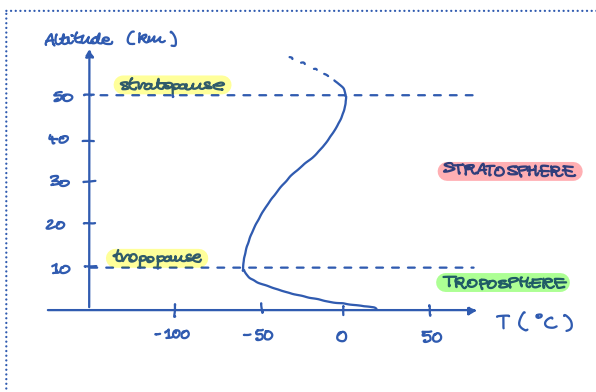
Dry atmosphere : $-10^\circ\text{C km}^{-1}$
 Moist atmosphere : $-5-6^\circ\text{C km}^{-1}$ } adiabatic T gradient

Ocean : $\frac{g\rho}{c^2} \sim 4.4 \times 10^{-3} \text{ kg m}^{-4}$, $c = 1520 \text{ m s}^{-1}$ } adiabatic density gradient

$\hookrightarrow$ Potential density increases with depth at smaller rate $0.6 \times 10^{-3} \text{ kg m}^{-4}$.

$\hookrightarrow$ Most in situ density increase due to compressibility and not due to T/salinity.

$\hookrightarrow$ Take into compressibility effects by reference density measurements to sea-level pressure



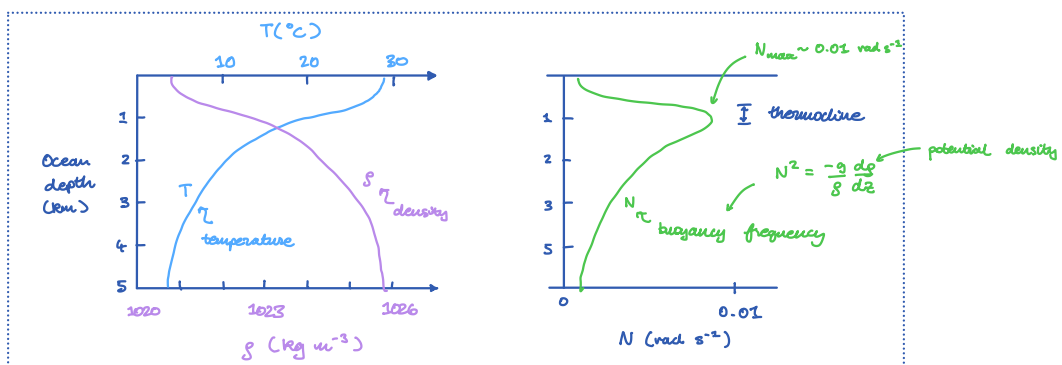
Troposphere : (10–15 km in height)

- Temperature decreases at $6.5^\circ\text{C km}^{-1}$
 - $\hookrightarrow$ Close to moist adiabatic lapse rate
 - $\hookrightarrow$ Close to neutral stability
 - $\hookrightarrow$ Expected due to turbulent mixing from frictional & convective effects for well-stirred atmosphere
 - $\hookrightarrow$ Close to neutral stratification. ($\frac{\partial \rho_0}{\partial z} = 0$)
- Clouds, weather changes, water vapour, ...

Tropopause : Average height $\sim 10 \text{ km}$ ($\sim 8-9 \text{ km}$ above poles, $> 15 \text{ km}$ above equator)

Stratosphere : Temperature increase due to UV absorption by ozone.
 • Stably stratified
 $\hookrightarrow$ Inhibits mixing & caps off turbulence & convective motion in troposphere.

Stratopause : $\sim 50 \text{ km}$ at all latitudes.



- Majority of the temperature increase with height due to absorption of solar radiation in upper layer of ocean.
- Density distribution in oceans affected by salinity but only secondary effect
 - No characteristic variation of salinity with depth
- Upper 50 - 200 m of ocean relatively well-mixed
 - Turbulence from wind, waves, current shear
 - Convective overturning due to surface cooling
- Small temperature gradients small below 1500m
 - Large " " " " large at 100 - 500 m. $\rightarrow$ high stability layer is "thermocline".

13.3 Equations of motion

For a stratified fluid, constant angular velocity Ω :

$$\nabla \cdot \mathbf{u} = 0, \quad \frac{D\rho}{Dt} = 0, \quad \frac{D\mathbf{u}}{Dt} + 2\boldsymbol{\Omega} \times \mathbf{u} = -\frac{1}{\rho_0} (\nabla p + \rho g \mathbf{e}_z) + \mathbf{F}$$

$\frac{D\mathbf{u}}{Dt}$ $\xrightarrow{\text{acceleration}}$ $\frac{1}{\rho_0} \nabla p$ $\xrightarrow{\text{pressure}}$ $\rho g \mathbf{e}_z$ $\xrightarrow{\text{gravity}}$ $\mathbf{F}$ $\xrightarrow{\text{body force}}$

diffusive effects omitted

body force (centrifugal force absorbed)
friction force per unit mass

- (Boussinesq approximation:)
- density variations neglected everywhere except in gravity term
 - vertical lengthscale $\ll$ "scale height" of medium c^2/g
 - good for ocean $c^2/g \sim 200 \text{ km}$
 - but less good for atmosphere $c^2/g \sim 10 \text{ km}$

For closed set of equations, $\mathbf{F}$ must be appropriately related to (average) velocity field.

- Geophysical flows are commonly
- turbulent & anisotropic
 - horizontal $\gg$ vertical velocities.

Friction force $\mathbf{F}_i = \frac{\partial \tau_{ij}}{\partial x_j}$, τ_{ij} viscous stress

Large-scale geophysical flows $\rightarrow$ $\mathbf{F}$ typically provided by turbulent momentum exchange (viscosity negligible)

$\hookrightarrow$ complexity of turbulence $\rightsquigarrow$ impossible to relate $\mathbf{F}$ and (average) velocity field in simple way.

To proceed in rudimentary fashion including anisotropy, use eddy viscosity hypothesis & turbulent viscosity presumed to have directional dependence.

$\hookrightarrow$ Geophysical fluid media typically stratified $\rightarrow$ inhibits vertical momentum transport.

13.4 Geostrophic flow

$\rightarrow$ Assume quasi-steady, large scale motion away from boundary

$p' \rightarrow$ pressure perturbation

$$\begin{aligned} -f v &= -\frac{1}{\rho_0} \frac{\partial p'}{\partial x} = -\frac{1}{\rho_0} \frac{\partial p}{\partial x} \\ f u &= -\frac{1}{\rho_0} \frac{\partial p'}{\partial y} = -\frac{1}{\rho_0} \frac{\partial p}{\partial y} \end{aligned}$$

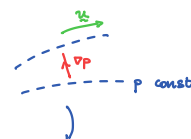
Horizontal equilibrium is geostrophic balance:

Reynolds number = $\frac{\text{Nonlinear acceleration}}{\text{Coriolis}} \sim \frac{u^2/L}{fL} = \frac{u}{fL} = Ro \ll 1$ for ocean. $\rightarrow$ can neglect nonlinear terms ($\mathbf{u} \cdot \nabla \mathbf{u}$)

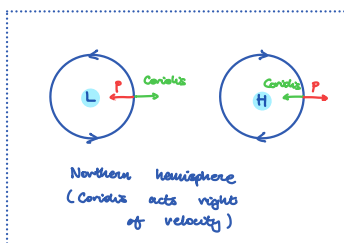
- $\hookrightarrow$ Equilibrium breaks down
- near equator (within $\pm 3^\circ$ latitude belt) with f small
 - friction/unsteadiness becomes important.

Note:

- $\mathbf{u} \cdot \nabla p = 0 \Rightarrow \mathbf{u}$ along pressure contours
- We can use $\frac{p}{f\rho_0}$ as a streamfunction



Since friction becomes important at lower levels of atmosphere, have partial flow across isobars.



Thermal wind

In the presence of horizontal density gradient, geostrophic velocity develops vertical shear.

↳ Geostrophic & hydrostatic balances by differentiating geostrophic balance w.r.t. z & $\frac{\partial p}{\partial z} = -\rho g$:

$$-f \frac{\partial u}{\partial z} = -\frac{1}{\rho_0} \frac{\partial}{\partial z} \left(\frac{\partial p}{\partial z} \right) \Rightarrow \begin{cases} \frac{\partial u}{\partial z} = -\frac{g}{f \rho_0} \frac{\partial \rho}{\partial z} \\ \frac{\partial v}{\partial z} = -\frac{g}{f \rho_0} \frac{\partial \rho}{\partial y} \end{cases} \quad \text{Thermal wind equations}$$

↳ Baroclinic phenomenon ($p = \text{const.}$, $\rho = \text{const.}$ do not coincide)

Note: Barotropic fluid: $p = \text{const.}$, $\rho = \text{const.}$
 Baroclinic fluid: $\hookrightarrow$ intersecting.

13.5

Ekman layers

Previously, low Rossby number, steady, frictionless.

↳ Extend this analysis to within frictional layers developing on horizontal surfaces.

For viscous flows with no Coriolis & ∇p , friction only balanced by unsteady acceleration or advective $u \cdot \nabla u$.

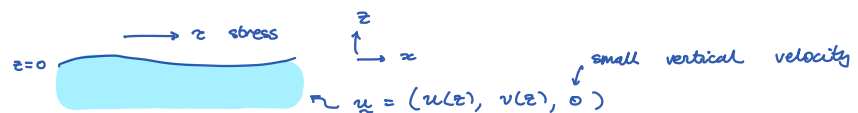
- Balance with $\frac{\partial u}{\partial t} \rightsquigarrow$ viscous layer which evolves with time (c.f. moving flat plate).
- Balance with $u \cdot \nabla u \rightsquigarrow$ boundary layer with thickness which could vary with direction.

In steadily rotating coordinate system, friction can be balanced by Coriolis.

↳ We have Ekman layers.

↳ Can be invariant in time and space.

Ekman layer at free surface

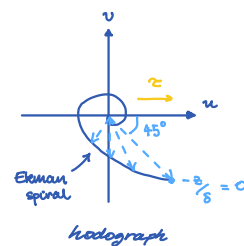


For simplicity, consider steady flow.

$$-f v = \nu \frac{d^2 u}{dz^2}, \quad f u = \nu \frac{d^2 v}{dz^2}$$

Boundary conditions: $\nu \frac{du}{dz} = \tau$ @ $z=0$ (stress in x at surface)
 $\frac{dv}{dz} = 0$ @ $z=0$ (no y -stress at surface)
 $u, v \rightarrow 0$ as $z \rightarrow -\infty$

Complex superposition: $V = u + i v$ complex velocity
 $\Rightarrow \frac{d^2 V}{dz^2} = \frac{i f}{\nu} V$ (Noting $\sqrt{-1} = \frac{1+i}{\sqrt{2}}$)
 $\Rightarrow V = A \exp \left\{ (1+i) \frac{z}{\delta} \right\} + B \exp \left\{ -(1+i) \frac{z}{\delta} \right\}$
 ($\delta = \sqrt{2 \nu / f}$)
 e-folding length



Volume transport: $\int_{-\infty}^0 u \, dz = 0$, $\int_{-\infty}^0 v \, dz = -\frac{\tau}{f}$
 ↳ net transport to right of τ & independent of ν
 ↳ Makes sense to balance effect of Coriolis acceleration with wind stress.
 ↳ also from $-f \delta v = \frac{d\tau}{dz}$ independent of eddy viscosity assumption.

Explanation with vortex tilting

In rotational flow, thickness of a viscous layer grows with time or downstream distance.
 In contrast, Ekman soln does not grow in time or space.

Vorticity components: $\omega_x = \frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} = -\frac{dv}{dz}$ (with $w=0$)
 $\omega_y = \frac{\partial w}{\partial z} - \frac{\partial u}{\partial x} = \frac{du}{dz}$

$\Rightarrow -f \frac{dv}{dz} = \nu \frac{d^2 \omega_y}{dz^2}$, $-f \frac{du}{dz} = \nu \frac{d^2 \omega_x}{dz^2}$
 ↳ vortex tilting, ↳ diffusion of vorticity

Without Coriolis, we would have thickening, but presence of planetary rotation means vertical fluid lines coincide with planetary vortex lines.
Tilting of vortex lines then cause rate of change of horizontal component of vorticity to cancel diffusion.

Ekman layers on rigid surface

Consider steady viscous layer on solid surface independent of x, y .
(e.g. atmosphere boundary layer over ground developing from winds above,
oceanic boundary layer at ocean bottom developing underneath uniform current in column)

Assume fluid velocity $U \hat{x}$ far from surface.
↳ Coriolis only balanced by pressure gradient, and $fU = -\frac{1}{\rho} \frac{\partial p}{\partial y}$
(geostrophic balance)
↑
geostrophic velocity

For $U, f > 0$ (Northern hemisphere), $\frac{\partial p}{\partial y} < 0$ so pressure force in +ve y -direction
↳ U is right of ∇p

Near solid surface, friction important in boundary layer:

$$-fv = \tau_v \frac{d^2 u}{dz^2}, \quad fu = \tau_v \frac{d^2 v}{dz^2} + \delta U$$

↑ replacing $-\frac{1}{\rho} \frac{\partial p}{\partial y}$

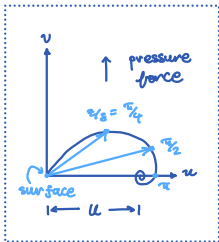
Boundary conditions: $u = U, v = 0$ as $z \rightarrow +\infty$
 $u = v = 0$ @ $z = 0$

$$\Rightarrow \frac{d^2 V}{dz^2} = \frac{2f}{\tau_v} (V - U) \quad \text{with} \quad V = U \quad \text{as} \quad z \rightarrow \infty$$

$V = 0$ @ $z = 0$

$$\Rightarrow V = U (1 - \exp\{-\frac{(1+i)z}{\delta}\}) \rightarrow \delta \text{ independent of } U.$$

↳ different to steady, non-rotating boundary layer on semi-infinite plate ($\delta \sim U^{-1/2}$)



Approaching wall, frictional effects decrease u and associated Coriolis acceleration.
↳ pressure gradient produces component v in direction of pressure force

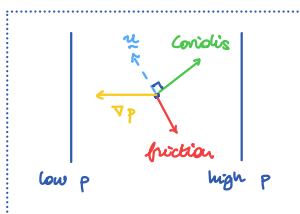
Net transport in Ekman layer normal to surface: $\int_0^\infty v dz = U (\frac{\tau_v}{2f})^{1/2} = \frac{1}{2} U \delta$
↑
left of free-stream velocity in direction of pressure force.

If atmosphere has laminar motion, $\tau_v = \text{molecular value for air}$

↳ Ekman layer at latitude 45° ($f \sim 10^{-4} \text{ s}^{-1}$) $\Rightarrow \delta \sim 0.4 \text{ km}$

↳ observed thickness $\sim 1 \text{ km} \Rightarrow \tau_v \sim 50 \text{ m}^2 \text{ s}^{-1}$

→ Demonstrates balance amongst Coriolis, pressure gradient & friction terms within boundary layer → results in component of flow towards lower pressure.



Note: u must have component towards low p when there is friction

→ Flow spirals inward to low pressure

↳ Mass conservation needs rising flow in low p system

↳ Cloud formation & rainfall → can also form cyclones

In high pressure system, air sinks and spirals out due to frictional effects

↳ clear skies & fair weather by suppressing cloud formation.

Friction effects (N.B. Ekman transport by surface winds) play major role in wind-driven ocean circulation.

Meeting 2 08/07/2026 (John, Bethan, Yuchen)

Papers to look at: Graye, Sarkar, Taylor (2009) Tidal/Stokes model
Taylor, Sarkar (2008) Stratified Ekman model

Summary: investigate effects of
1) Ekman (Coriolis, forcing to balance free-stream flow U)
2) Stokes (no Coriolis, no initial flow, oscillating ∇p)

To do: • Reproduce results from above papers.
↳ Later, see what happens with weak stratification for abyssal ocean.

Both simulations: ✓ drag on bottom boundary condition (c.f. Tilted BBL on Oceananigans)
✓ free slip at top
✓ buoyancy $b = -\frac{\rho}{\rho_0}$ b.c. → Nonhydrostatic model
✓ no flux at bottom.
✓ density gradient should match initial condition at top of domain.
✓ periodic in x, y (set via topology)
✓ initial condition with linear density profile
✓ add random noise to initial velocity for turbulence generation
✓ start 2D then 3D (for speed)
✓ use stretched grid (fine grid near bottom)
(see Tilted BBL on Oceananigans)
✓ use WENO advection scheme
↳ to change later → change back to Centred (default)
↳ use closure scheme later → Anisotropic Minimum Dissipation & Scalar Diffusivity

Ekman layer: ✓ Include Coriolis ($f = 10^{-4} \text{ s}^{-1}$)
✓ Initialise with const. uniform velocity U
✓ Add forcing
↳
$$\begin{aligned} \frac{\partial u}{\partial t} + \underline{u} \cdot \nabla u - f v &= -\frac{\partial p}{\partial x} \\ \frac{\partial v}{\partial t} + \underline{u} \cdot \nabla v + f u &= -\frac{\partial p}{\partial y} + f U \end{aligned}$$

Ekman layer height: $\delta = \sqrt{\frac{2\nu}{f}} = \frac{\sqrt{2}}{10} \sim 0.14 \text{ m}$
($\nu \sim 10^{-6}$, $f = 10^{-4}$)

In Taylor & Sarkar, simulate $tf = \mathcal{O}(10) \Rightarrow t = \mathcal{O}(10^5)$ with layer height $\sim 0.5\delta - 0.7\delta$

Revisit previous papers from Meeting 2

- To do:
- Update and change folder for Oceananigans
 - No-slip for Stokes problem
 - Change numerical inputs for each model [c.f. papers]

Taylor & Santhanam (2008): (Pentonic Ekman layer)

1 Introduction

- Stratification $\rightarrow$ suppression of turbulent production in boundary layer
- Unstratified Ekman layer has height $\approx 0.5 \frac{u_*}{f}$ e.g. midlatitude: $\left. \begin{array}{l} u_* = 1 \text{ cm s}^{-1} \\ f = 10^{-4} \text{ s}^{-1} \end{array} \right\} \Rightarrow h \approx 50 \text{ m}$
- In many cases (e.g. coastal environments) w/ large stratification
 - Mixed layer thickness of only 10m over Oregon shelf. (Perin et al. 2007)
 - Paper investigates change in Ekman layer structure
- Note: apply approximate drag boundary condition
 - Applying no-slip $\rightarrow \mathcal{O}(0.1 - 10 \text{ cm})$ viscous sublayer. (Caldwell & Chinn 1979)

$$\tau_w = \rho_0 \kappa (\tau_1 + \tau_2 \sqrt{u^2 + v^2})$$
 shear stress
 - Using quadratic drag, $\tau_1 = 0$, $\tau_2 = 2 \times 10^{-3} \Rightarrow \tau_w = c_d |u| u$ (Hauke Vogel & Beckmann 1999)
- Friction velocity: $u_* = \sqrt{\frac{\tau_w}{\rho_0}}$
 - Can use profile method (Johnson et al. 1994) + law-of-the-wall (e.g. Pope 2007)
 - Can fit with a logarithmic profile:

$$|u| = \frac{u_*}{\kappa} \ln\left(\frac{z}{z_0}\right)$$

$$\kappa \approx 0.41 \text{ von Karman}$$
 - Not valid through entire mixed layer.
- Assumption:
 - Flow outside boundary layer is steady, uniformly stratified & in geostrophic balance.
 - Get separation from mixed layer with strongly stable pycnocline

2 Formulation

- Assume:
 - Free stream in geostrophic balance
 - Seafloor is flat & rough.
 - Use near-wall model.
 - x, y periodic (flow statistically homogeneous in horizontal plane)
 - Domain large (flow decorrelated over distance equal to domain size)
 - Domain size $0.108D \times 0.108D \times 0.0405D = 72.8 \text{ m} \times 72.8 \text{ m} \times 27.3 \text{ m}$

Quantities: $U_{\infty} = 0.0674 \text{ m s}^{-1}$, $f = 10^{-4} \text{ s}^{-1} \Rightarrow D = \frac{U_{\infty}}{f} = 674 \text{ m}$
 $\hookrightarrow$ changed to 0.04 m s^{-1}

$$Re_{\infty} = \frac{U_{\infty} D}{\nu} \sim 4.55 \times 10^7$$

$$Ra_{\infty} = -\frac{g}{\rho_0} \frac{d\rho}{dz_{\infty}} \frac{D^2}{U_{\infty}^2}$$

$$Pr = \frac{\nu}{\kappa} \sim 10$$

TABLE 1. Relevant physical parameters. Re_{∞} , N_ω/f , Pr , and z_0 are input parameters and the other parameters are outputs of the numerical model.

Re_{∞}	Re_{ω}	Ri_{ω}	N_ω/f	Pr	u_d/U_{∞}	z_0/δ	α_0
4.55×10^7	1.08×10^5	0	0		0.0488	4.80×10^{-5}	15.4°
4.55×10^7	1.09×10^5	1000	31.6	5-10	0.0490	4.78×10^{-5}	18.9°
4.55×10^7	1.12×10^5	5625	75		0.0497	4.71×10^{-5}	24.8°

Bad assumption $\rightarrow z = 10^{-6} \text{ m}^2 \text{ s}^{-1}$, $\kappa = \frac{\nu}{Pr}$ Perin et al. (2005)
 $\hookrightarrow$ change to statistical later $\rightarrow$ Assume $\frac{u_*}{U_{\infty}} = 0.049 \Rightarrow \delta = \frac{u_*}{f} \approx 33 \text{ m} \rightarrow z_0 \approx 0.16 \text{ cm}$ (consistent with $0.05 - 2 \text{ cm}$)

3 Numerical method

- Uses a grid $128 \times 128 \times 201$, $72.8\text{m} \times 72.8\text{m} \times 29.3\text{m}$
- Horizontally $\rightarrow$ derivatives calculated with pseudospectral method
- Vertically $\rightarrow$ 2nd order finite differences
- Time-stepping: mixed explicit & implicit scheme (RK3, CN3)
 - $\hookrightarrow$ Ensures conservation mass, momentum, energy
- Low pass filter prevents formation of spurious energy near grid scale. (apply on u & T)
- 4th order compact filter with sharp wavenumber cutoff (lele 1992) applied vertically every 10 steps.
- Open boundary condition implemented at top of computational domain prevents spurious reflection of internal gravity waves.
 - $\hookrightarrow$ use sponge layer with radiation boundary condition $\rightarrow$ only 6% reflection.
- Subgrid-scale stress tensor τ & density flux $\tilde{\rho}$ evaluated using dynamic Smagorinsky ($C, C_s \geq 0$ plane-averaged)
- For accuracy of large eddy simulation (LES), we need to resolve turbulent scales responsible for substantial ($\sim > 50\%$) of turbulent kinetic energy.
- Local Ellison scale $L_E = \frac{\sqrt{\theta'}}{(d\theta'/dz)} \rightarrow$ Taylor & Sarkar (2008) shows that this provides reasonable estimate for scale of turbulent eddies

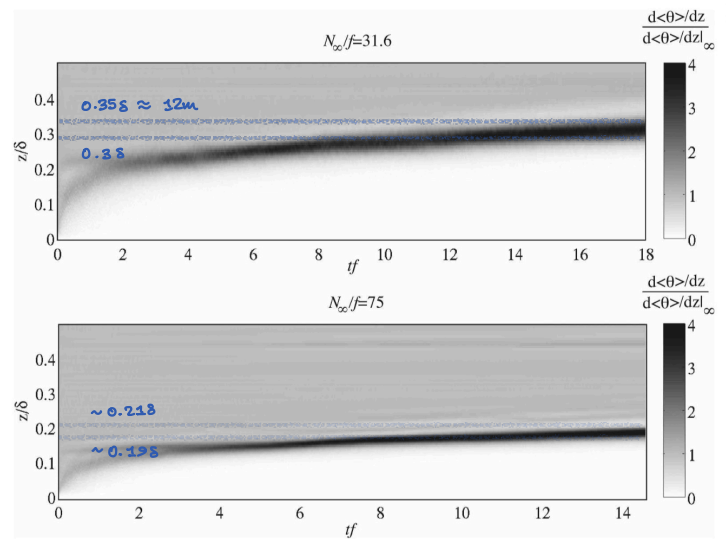
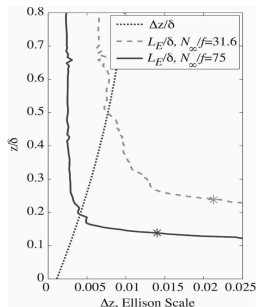
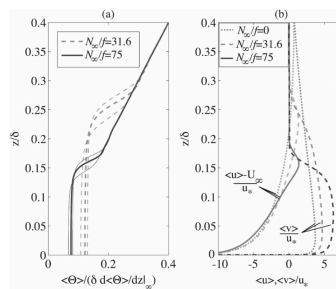


FIG. 3. Evolution of the plane-averaged temperature gradient.

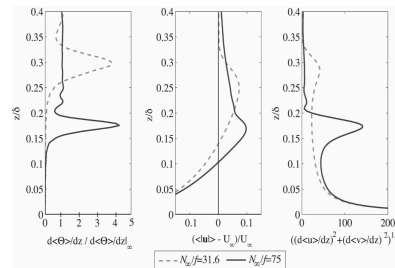


Ellison scale



Buoyancy profile

Velocity profile



Buoyancy gradient

Horizontal velocity

Mean shear

1 Introduction

2 Formulation

M_2 tidal period 12.4 h

Buoyancy frequency N_∞

Free stream

$$U_{\infty,d}(t_d) = U_{o,d} \sin(\omega_d t_d)$$

Pressure gradient

$$\frac{dp_d}{dx_d}(t_d) = -\rho_{o,d} U_{o,d} \omega_d \cos(\omega_d t_d)$$

$$\left[\begin{array}{l} \text{Navier-Stokes: } \rho \frac{\partial u}{\partial t} + \rho \mathbf{u} \cdot \nabla \mathbf{u} - f_v = -\frac{\partial p}{\partial x} = \rho U_o \omega \cos(\omega t) \\ \rho \frac{\partial v}{\partial t} + \rho \mathbf{v} \cdot \nabla \mathbf{v} + f_u = -\frac{\partial p}{\partial y} \end{array} \right]$$

- Neglect rotation of the Earth

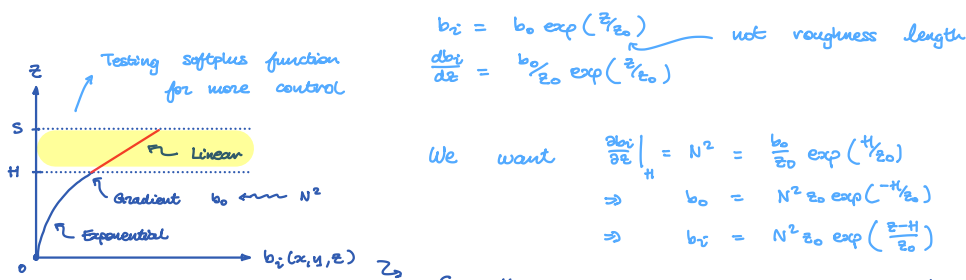
↳ Focus on stratification on turbulent velocity & thermal boundary layers

- To do:
- Both:
 - Plot velocity & vorticity profile over time (add to anim.jl)
 - Ekman:
 - implement sponge (relaxation in additional sponge layer $\vec{u} \rightarrow \vec{u}_w$ & $b/N^2 z \rightarrow 1$)
resolved! $\rightarrow$ check viscosity & diffusivity implemented by AMD closure $\rightarrow$ used Gaussian mask
 - Try exponential initial profile. $\rightarrow$ try softwise/piecewise linear later $\rightarrow$ may work for Smagorinsky

Taylor & Sarkar (2007) [Internal gravity waves generated by turbulent bottom Ekman layer]

2 Formulation

- Generated by steady flow in geostrophic balance encountering smooth flat wall.
- With friction velocity u_* and turbulent Ekman layer depth $\delta = \frac{u_*}{f}$, outer layer density gradient $\frac{d\rho}{dz_\infty}$:
 $Re_* = \frac{u_* \delta}{\nu}$, $Ri = \frac{-g}{f \delta_\infty} \frac{\delta^2}{u_*^2} = \frac{N_\infty^2}{f^2}$, $Pr = \frac{\nu}{\kappa}$ [all are parameters for the simulations]
- Change in density assumed to be caused due to temperature variation $\leadsto Pr = 5$
- Changing $Ri_* \rightarrow$ changing free-stream temperature gradient.
- [Perlin et al. (2007)] $\rightarrow Re_* \approx 60\,000$, $N_\infty/f = 75$



Currently, all exponential initial buoyancy profile, with Linear Target in Relaxation.

- Think this is likely problematic

$\rightarrow$ Stick to Relaxation with "target" = $(x, y, z, t) \rightarrow b_i(x, y, z)$

* hindsight: have tried various profiles in week after

↳ $b_i = N^2 [z + \ell(e^{-z/\ell} - 1)] \rightarrow$ Change to $b_i = N^2 \ell [1 - \exp(-\frac{z}{\ell})]$
 Match Ekman & Stokes model parameters $\uparrow n=5$ at the moment

↳ $U_0 = U_\infty = 0.04 \text{ m s}^{-1} \rightarrow \text{Change } Re_\infty = \frac{U_\infty D}{\nu_0}$ accordingly

$$\left. \begin{aligned} \hookrightarrow P_r &= 10 = \frac{v}{\kappa} \\ \hookrightarrow v &= 10^{-6} \text{ m s}^{-1} \end{aligned} \right\} \Rightarrow \kappa = \frac{v}{P_r} = 10^{-7}$$

$$\hookrightarrow \gamma = 10^{-6} \text{ m}^2 \text{ s}^{-1}$$

↳ likely too complicated after John's feedback.

↳ load into profiles 3 2 4.

(my suggestion)

point for sharp change at $z = T$

+ satisfies $\frac{\partial b}{\partial c} = 0$ @ $c = 0$

- does decay to linear profile, but slow

(my suggestion)

$$+ \frac{2}{3} = 0 \quad \text{②} \quad u = 0$$

+ much faster decay to linear profile

- may be too complicated

- not much evolution in initial profile.

(previously considered)

- + Reaches a linear profile quickly

- Wide variation only for small decay lengths L

(suggestion by John
w/ corrections,
similar to Yuchen's)

+ Fixed buoyancy change $\Delta b = b(t) - b(0)$

- Limit as $L \rightarrow \infty$ is $b_{\text{lin}}(z) = \Delta b \left(\frac{z}{H} \right)^2$
(quadratic profile)

- For profile 4, _____ " _____ $l =$

[perhaps play around with sponge_rate in sponge layer relaxation]

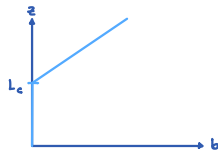
$$\text{sponge_rate} = \text{const} \times N = \text{const} \times \text{fr}$$

Ratio $r := \frac{N}{f}$

Profile 5: $b_i(z) = \frac{N^2}{S} \ln [1 + e^{S(z-T)}]$

Things to check on: • use of profiles 3 & 4

To do: ✓ try a piecewise linear profile



$$b_i(z) = \begin{cases} N^2 z & , z > L_c \\ 0 & , 0 < z < L_c \end{cases}$$

→ can smooth this out using softplus function

$$b_i(z) = \frac{N^2}{S} \ln[1 + e^{S(z-L_c)}]$$

J • try simulation with $N=0$ whilst having $b_i \rightarrow$ use as control

↳ # out "buoyancy = :b" in "model" → acts as a passive dye

B ? use data about the ocean → can try at a different stage

J ✓ King et al. (2012)

↳ details buoyancy profiles → $N^2 \sim 10^{-8} \text{ rad}^2 \text{ s}^{-2}$

↑ ω — typical frequency of internal tides (internal gravitational waves)

↳ $\omega \in [f, N]$

↳ if $N < f$, then this doesn't occur (turning depths: depths at which this occurs)

✓ Set height $L_z = 40 \rightarrow$ potentially increase for greater Ekman layer

Y • New setup skips initial slow mixed-layer growth & reaches equilibrium faster

↳ Can set T to be closer to layer-depth value

• Skips initial pycnocline formation

? What is the maximum mixed-layer depth?

↳ Test with $r=0$ for classical Ekman layer height $\delta = \frac{u_*}{f}$

↳ Armi & Millard suggest 6.5

• Output bottom boundary stress to calculate δ more accurately

Turbulent kinetic energy (TKE): $\langle k \rangle_{xy} = \left\langle \frac{1}{2} \sum_i \overline{(u_i')^2} \right\rangle_{xy}$

- $u' = u - \bar{u}$
 $\bar{u}$ horizontally-averaged & time-averaged

- Make TKE plot with depth z

↳ Want to use TKE to analyse decay lengthscale

- Two mixing phases:

↳ Initially fast

↳ Slow mixing: after initial layer forms
 KE $\Rightarrow$ PE

- Can use TKE to calculate PE

- Quantify how TKE may decay as a function of depth for same stratification

↳ can predict rate of change of PE

- Calculate rate of change of PE, mean PE, mean horizontally-averaged ρ profile, KE as function of time.

→ Look at ε = dissipation of TKE.

- Considered timescale of dense water moving by currents may be relevant to mixing.

- Look at diffusivity K as function of layer height [Taylor & Sarker or
 relatively unaffected by outer-layer strat. $\hookrightarrow u_* = (\varepsilon K z)^{1/3}$]

- Extract • model closure - fields

- moments $uu, vv, ww,$
 uv, vw, uw
 wb

$$K_T = h^{\alpha} \sqrt{\overline{\rho \varepsilon}} \cdot f$$

$$\log K_T = \alpha \log h + C$$

$$[K_T] = m^2 s^{-1}$$

$$[u] = m s^{-1}$$

$$[T] = s$$

$$[N] = s^{-1}$$

Tennekes & Lumley:

2 Turbulent transport of momentum & heat

3 Dynamics of turbulence

3.1 Kinetic energy of mean flow

From 2.1, steady mean flow & incompressible:

$$u_j \frac{\partial u_i}{\partial x_j} = \frac{\partial}{\partial x_j} \left(\frac{\tau_{ij}}{\rho} \right), \quad \frac{\partial u_i}{\partial x_i} = 0$$

Since $\bar{u}_i = 0$ for turbulent velocity fluctuations, so look at KE.

$$\tau_{ij} = \frac{1}{2} \rho u_i u_j$$

$$\tau_{ij} = -P \delta_{ij} + 2\mu S_{ij} - \rho \overline{u_i u_j}$$

$$\begin{cases} PE = -\langle z b \rangle \\ KE = \frac{1}{2} \langle |u|^2 \rangle \end{cases} \quad \begin{cases} \frac{\partial}{\partial t} PE = -\langle w'b' \rangle \\ \frac{\partial}{\partial t} KE = +\langle w'b' \rangle - \varepsilon + P \\ \frac{\partial}{\partial t} TKE = -\langle w'b' \rangle - \varepsilon \end{cases} \quad \begin{matrix} \nearrow \\ MKE + TKE = KE \end{matrix}$$

$$\overset{\text{SAS}}{K_T} \frac{\partial b}{\partial z} + w'b' = -K_T \frac{\partial b}{\partial z}$$

$$\begin{aligned} \langle w'b' \rangle &= \frac{1}{L_z} \int_0^z w'b' dz \rightarrow \text{can use directly} \\ &= \frac{1}{L_z} \int K_T \frac{\partial b}{\partial z} dz \\ &\approx \frac{1}{L_z} K_T \end{aligned}$$

Goal: 1) Find TKE as some function of z

↳ For Ekman, profile seems universal but seems like

$\text{TKE}(z) \propto e^{-\alpha z} \Rightarrow \log(\text{TKE}) \propto (-)z$ (and rescale $\text{TKE} \rightarrow \frac{\text{TKE}}{u_{\infty}^2}$ so it applies for

2) Find turbulent diffusivity $K_T(\text{TKE}, N) \rightarrow$ depends on stratification other velocity profiles)

↳ Seems that $K_T \sim \frac{\text{TKE}}{N}$ for large Ri (turbulence suppressed by stratification) $\rightarrow$ 

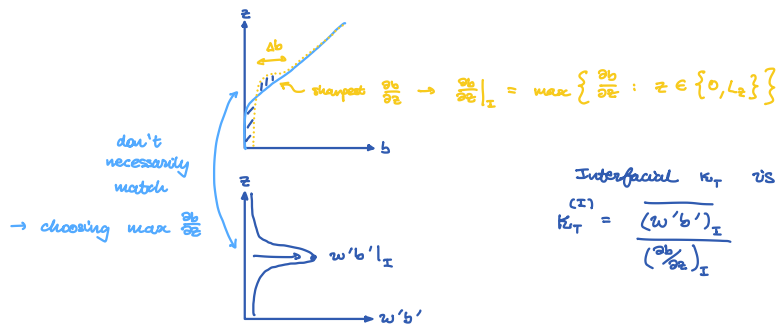
$K_T \sim L\sqrt{\text{TKE}}$ for small Ri (turbulence suppressed by layer height) $\rightarrow$ 

[units of $[K_T] = \text{m}^2 \text{s}^{-1}$]

↳ In this case, we can test $L = \begin{cases} \frac{u_*}{S} \\ \frac{u_*}{\omega} \end{cases}$ OR use actual layer height from simulation

↳ Numerical implementation:

• Interested in interfacial value of K_T



Interfacial K_T is given by

$$K_T^{(I)} = \frac{\overline{(w'b')_I}}{\left(\frac{\partial b}{\partial z}\right)_I}$$

To do:

- Ekman: • calculate $K_T^{(I)}$
- Stokes: • calculate $\text{TKE}(z)$

Meeting 9

25/08/26

(John, Bethan, Yuchen)

To do: • Look at shear lengthscale $l_s = \frac{\sqrt{TKE}}{(\partial \bar{u} / \partial z)_I}$

J ✓ Useful to have correlation FMC & error bars instead of scatter

- Gradient Richardson often used as parameter

$$l_N = \frac{K_0}{\sqrt{TKE}}$$

$$l_N = \frac{\sqrt{TKE}}{N}$$

$$l_S = \frac{\sqrt{TKE}}{|\partial u / \partial z|}$$

B ◦ Investigate $\frac{N^2}{(\partial u / \partial z)^2}$ as function of $r = N/g$

Y ◦ Plot $\frac{K_T}{TKE}$ vs. $(\partial u / \partial z)^{-1}$ timescale

J ◦ l_S vs l_N

↳ caveat → may have coefficient

Y ◦ See how lengthscales vary with r

J ◦ Functional form (exp) of l_N with l_N vs. l_S → shows structure with decreasing r .

- Theory of l_N vs. l_S

- $l_c = \frac{1}{(\frac{C_N}{l_N} + \frac{C_S}{l_S})}$ (weighted harmonic mean)

$$\frac{z}{(1 + N^2/g^2)^{0.25}}$$

- Predict $TKE(z) \longleftrightarrow K_T$

- Normalise height $z/s \leftarrow$ modified depth in Weatherly (dependent on r)

$$\log\left(\frac{TKE}{u^2}\right) = \frac{mz}{h_0} + c$$

$$TKE = Ae^{-mz/h_0}$$